

Air Pollution, Water Pollution and Radioactive Pollution

AIR POLLUTION

Air pollution is defined as the composition of air is disturbed due to presence of gas, smoke, dust, chemical particulate etc., which are injurious to human being, plant and animal.

Types of Air pollution

- i) **Primary pollutant:** Pollutants that are emitted directly from the human or natural activities are known as primary pollutant. For example, CO_2 , SO_2 , NO_x , particulate matter, hydrocarbons etc.
- ii) **Secondary pollutant:** when primary pollutants are react with atmospheric moisture content then a new category of pollutants is form, known as secondary pollutant. For example, carbonic acid, nitric acid, sulphuric acid etc.

Cause of Air pollution

- a) Urbanization
- b) Population
- c) Deforestation
- d) Industrialization
- e) Vehicle emission

Major Air pollutants and their effects

- a) Carbon dioxide: the concentration of CO_2 gas increase in atmosphere due to emission from vehicles, burning of fossil fuel, emission from volcano, industries, agricultural activity etc. it increase green house effect which causes global warming and climate change.
- b) Carbon monoxide: carbon monoxide gas releases after incomplete combustion of fossil fuel or other product. The source of CO is vehicle emission, burning of coal, biomass combustion etc. CO causes headache, dizziness, heart failure (in blood CO combines with oxygen which reduced the affinity of haemoglobin towards oxygen), etc.
- c) Sulphur dioxide: SO_2 releases from oil refineries, volcanic eruption, and chemical industries etc. sulphur dioxide react with moisture to form

secondary pollutant which causes eye irritation. It can also cause allergic reaction and asthma.

- d) Lead: tetra ethyl lead used as anti-knocking agents in petrol for smooth function vehicle. Lead particle coming out from the exhaust of vehicle and mixed with air. It causes injurious effect on kidney and liver. It also lowers down the intelligence power in children.
- e) Nitrogen oxide: it release from vehicle exhausts, volcanic eruption, lighting etc. like SO_2 , it also react with moisture content present in atmosphere and causes eye irritation.

Techniques used for prevention of Air pollution

- i) Filters: filter remove particulate matter from the gas stream. Bag house filter system is the most common and it is made up of cotton fibers. When polluted gas passed through it, then polluted gas are deposited on cotton fibers.
- ii) Electrostatic scrubber: the emitting dust is charged with ions and ionized particulate matter is collected on oppositely charged surface. The collected particles are removed by shaking the surface.
- iii) Scrubbers: Scrubbers are wet collectors. They remove aerosol from a stream of gas either by collecting wet particle on a surface followed by their removal on the particles are wetted by scrubber liquid.

WATER POLLUTION

The undesirable biological or chemical substance present in water which adversely affect living organism is referred as water pollution.

Source of Water pollution

- i) **Point source pollution:** if pollutants discharge from single identifiable source then it is known as point source. For example, municipal sources, industrial sources etc.

- ii) **Non-point source pollution:** if pollutants discharge from random or scattered source then it is known as non-point source. For example, construction site, agricultural site, Acid rain, animal waste etc.

Cause & effects of Water pollution

- a) Disease causing agents: the micro-organism including bacteria, virus, protozoa, if present in drinking water causes disease. For example,
- b) Oxygen depleting waste: organic matters present in water are degraded by microorganism present in water which required oxygen. If large amount of organic matter present in waste water then large amount of oxygen is required by the microorganism to degrade the waste. Therefore oxygen content in water decrease. The amount of oxygen consumed by microorganism is referred as Biological Oxygen Demand (BOD). High level of BOC means large amount of waste present in water.
- c) Water soluble inorganic chemicals: the elements like lead, mercury, cadmium, arsenic adversely affect the human being and animals. For example, cadmium causes Itai-Itai disease, mercury causes Minamata disease.
- d) Suspended solids: if suspended solid present in water bodies, then water become turbid and therefore proper sunlight does not reach to the aquatic plant and animal which disturb the life of aquatic ecosystem.

Control measure for preventing water pollution

- i) Industrial effluent and domestic waste must be treated before disposal.
- ii) Recycling of waste water through waste water treatment.
- iii) Public awareness program.

Radioactive Pollution:

What is Radioactive Pollution?

- When radioactive materials are present or added to the environment, especially when doing so unintentionally and endangering the environment through radioactive decay, this is known as radioactive pollution.
- The radioactive materials cause damage by releasing dangerous ionizing radiation into the environment where they are present, such as beta or alpha particles, gamma rays, or neutrons.
- About 20% of the radiation humans are exposed to is thought to be caused by human activity.
- Human activities that can release radiation include those involving radioactive materials, such as mining, handling and processing radioactive materials, handling and storing radioactive waste, using radioactive reactions to produce energy, and using radiation in research and medicine.

Radioactivity

- Radioactivity is the term used to describe the spontaneous emission of particles or waves from the unstable nucleus of some materials. The three different categories of radioactive particles are alpha, beta, and gamma.
- Alpha particles are those that have a positive charge. Gamma rays are neutral electromagnetic radiations, whereas beta particles are negatively charged electrons.
- Natural radioactive materials exist in the crust of the world.
- Uranium, thorium, and actinium are three NORM (Naturally Occurring Radioactive Materials) series that damage water supplies.

Sources of Radioactive Pollution

Exposure to Cosmic Radiation

- The outer atmosphere of the earth is continually being attacked by cosmic radiation.
- Fast-moving particles that are present in space and can originate from a variety of sources, including the sun and other celestial events, make up cosmic radiation.
- Cosmic rays are mostly protons, but they can also be other particles or wave energy.

Terrestrial Radiation

- The Earth itself is the source of terrestrial radiation. Soil and rock can both contain naturally occurring radioactive elements.
- The main sources are uranium, potassium, and thorium natural deposits, which during natural decay generate small amounts of ionizing radiation.
- Both uranium and thorium are “ubiquitous,” or present practically everywhere.

Radiation through Inhalation

- The majority of fluctuations in natural radiation exposure are caused by the inhalation of radioactive gases produced by radioactive elements found in soil and bedrock.
- Radon, a radioactive gas that has neither an odor nor a color, is created when uranium-238 decays. Because it is an inert gas, it does not interact with its surroundings.
- Radon does not react, thus it can move readily through the soil and up into the atmosphere.
- A radioactive gas generated from thorium is called thoron.
- The composition of the soil and bedrock has a significant impact on the amount of radon and thoron in the air.

Nuclear Power Plants

- The primary cause of the production of radioactive waste is the nuclear fission process in nuclear power plants.
- These operations produce radioactive wastes that are a significant environmental risk connected with nuclear power facilities, including spent (used) reactor fuel, uranium mill tailings, and other radioactive wastes.
- These substances can remain radioactive for a very long time and are dangerous to human health.

Nuclear Waste Handling and Disposal

- Low to medium quantities of radioactivity can be produced over time through the treatment and disposal of nuclear waste.
- The air, water, and soil could all become contaminated by the radiation.
As a result, it could be challenging to recognize and foresee their effects. Additionally, certain nuclear waste sites might go unnoticed.
- The main issue in handling radioactive waste is that it cannot be handled biologically or chemically.

Nuclear Weapons

- With the start of the atomic age came the first nuclear weapons tests, which led to the radioactive contamination of many areas all over the planet.
- The Stockholm International Peace Research Institute platform has statistics showing that 2053 nuclear tests were carried out worldwide between 1945 and 2006.
- One of the main causes of radioactive pollution was this.

Effects of Radioactive Pollution

Genetic Mutations

- Radiation has detrimental effects on DNA and genetics.
- Genetic decay results from the harm it causes to DNA strands over time.
- Depending on the quantity and type of radiation one has been exposed to, the degree of genetic mutation resulting in changes in DNA composition varies.
- A person or animal exposed to excessive radiation through the environment, food, or even water is likely to have already absorbed the radiation into their systems.
- Energy cannot be expelled, thus once within the body, it stays active.
One is very prone to cancer as a result of the mutation.

Diseases

- Radioactive pollution has a negative impact on our health. One of the rarest and deadliest side effects of radioactive pollution is acute radiation sickness.
- However, it is a result of radioactive radiation at high levels. This condition quickly results in nausea and vomiting.
- In the worst-case scenario, the person can pass away within a few days or weeks.
- Cancer is the most frequent adverse reaction to radiation and it can also cause it.

Soil Infertility

- Because soils are exposed to the atmosphere, radiation is a common occurrence in them.
- The soil's radioactive chemicals react with different nutrients, destroying the nutrients and making the soil poisonous and infertile.
- Such soil causes the harvest of radioactively tainted crops that are dangerous for ingestion by both humans and animals.

Impact on Aquatic Life

- As a source of nuclear energy and chemical processing, power plants have been discharging radioisotopes into the water for many years.
- To mention a few, these include Cesium, Radon, Crypton, Ruthenium, Zinc, and Copper.
- The trash is not necessarily safe just because it is being released at a "permissible" level.
- Both the soft tissues and the bones of the fish contain these radionuclides.
- It was thought that the seaweed used in bread contained the ruthenium radioisotope
- Radionuclides are present in the tissues and shells of all fish with shells.

Measures to Control Radioactive Pollution

Proper Method of Disposing of Radioactive Waste

- Radioactive waste still contains radiation. It cannot be disposed of in the same way as conventional waste because of this.
- It can neither be buried nor burned. This trash should be kept in big, sturdy concrete containers because seepage is a possibility.
- Another option is to dilute the radiation because storage might not be practical.

Proper Labeling

- Any product containing radioactive material must be labeled, and the label's content must include the necessary safety instructions.
- This is because radiation can enter the body through very slight interaction with radioactive material.
- To encourage the use of protective equipment when handling them, containers containing hazardous products must to be prominently marked.

Prohibition of Nuclear Tests

- It has previously been proven that nuclear energy has a significant amount of latent destructive power.
- However, the tests made to improve the energy have a big impact on the total amount of radioactive elements.
- These tests also end up leaking into adjacent ecosystems despite being done in deserts, affecting many people's livelihoods.

Alternative Energy Sources

- Nuclear power development and use were initially not bad things.
- But given the harm and dangers it does to the environment, it is long past due for its use to be phased out in favor of more eco-friendly energy sources like renewable energy (solar, hydroelectric, and wind power).
- The waste released from the various processes and combustion causes additional radiation to be released into the atmosphere when radioactivity is used to generate energy in nuclear power plants, for instance.